

Determinant of the $i+j$ Matrix

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Goal

Let $n \in \mathbb{Z}_{\geq 1}$ and define the $n \times n$ matrix

$$A = (A_{ij})_{1 \leq i, j \leq n}, \quad A_{ij} = f(i, j).$$

In this document we take

$$f(i, j) = i + j,$$

so $A_{ij} = i + j$. Prove that $\det(A) \equiv 0 \pmod{n}$ (and in fact $\det(A) = 0$) for all $n \geq 2$.

Example: the 3×3 case

For $n = 3$,

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}.$$

Apply row operations that do not change the determinant: replace a row by itself minus another row.

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - R_2 \text{ (using the new } R_2\text{)}.$$

After the first operation,

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 1 & 1 \\ 4 & 5 & 6 \end{bmatrix}.$$

After the second operation,

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 1 & 1 \\ 3 & 4 & 5 \end{bmatrix} \quad \text{and now do } R_3 \leftarrow R_3 - R_1,$$

giving

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

This matrix has two identical rows ($R_2 = R_3$), so its determinant is 0. Since each step was a determinant-preserving row operation, the original matrix also has determinant 0:

$$\det(A) = 0.$$

Therefore $\det(A) \equiv 0 \pmod{3}$.

General proof for all $n \geq 2$

Let $n \geq 2$ and let A be the $n \times n$ matrix with $A_{ij} = i + j$.

Step 1: Express columns in a low-dimensional span

Fix a column index j . The j th column of A is the vector

$$C_j = \begin{bmatrix} 1+j \\ 2+j \\ \vdots \\ n+j \end{bmatrix}.$$

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ n \end{bmatrix}.$$

Then

$$C_j = j \mathbf{u} + \mathbf{v}.$$

So every column of A lies in the span of $\{\mathbf{u}, \mathbf{v}\}$, which has dimension at most 2.

Step 2: Conclude the determinant is zero when $n \geq 3$

If $n \geq 3$, then A has n columns, all contained in a subspace of dimension ≤ 2 . Hence the columns are linearly dependent, so $\det(A) = 0$.

Step 3: Handle the $n = 2$ case directly

When $n = 2$,

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}, \quad \det(A) = 2 \cdot 4 - 3 \cdot 3 = 8 - 9 = -1 \neq 0.$$

Wait: this shows $\det(A) \neq 0$, so we must re-check the general argument.

Correction: the entry pattern is $i + j$, but indexing matters

In our definition, $A_{ij} = i + j$ with $i, j \in \{1, \dots, n\}$. For $n = 2$, the matrix is indeed $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ and its determinant is -1 , not 0 .

So the correct general statement is:

$$\det(A) = 0 \text{ for all } n \geq 3,$$

and for $n = 1, 2$ the determinant is nonzero.

Step 4: Modular conclusion

For every $n \geq 3$, $\det(A) = 0$ as an integer, hence in particular

$$\det(A) \equiv 0 \pmod{n}.$$

Summary

For $A_{ij} = i + j$:

$$\det(A) = \begin{cases} 2, & n = 1, \\ -1, & n = 2, \\ 0, & n \geq 3. \end{cases}$$

Therefore $\det(A) \equiv 0 \pmod{n}$ for all $n \geq 3$.